Oct 11th, 2026 Editor: Advaith K High: 65°F Low: 56°F

News Weekly

Taking antibiotics? Sugar may be making the damage worse

New research suggests that consuming more sugar while on antibiotics may worsen the disruption these drugs cause to beneficial gut bacteria. The study examined the intestinal microbiome, which assists in digestion, immune function, and other vital processes. Antibiotics, while targeting harmful bacteria, also affect beneficial ones, potentially allowing dangerous species to thrive. A diverse microbiome can resist harmful pathogens, but antibiotics may reduce this diversity, increasing disease vulnerability. Researchers from NYU Langone Health and others monitored 173 cancer patients' meals and found a 21% decline in beneficial bacterial diversity for every 100 grams of sugar consumed before treatment. Sugary diets combined with antibiotics increased Enterococcus faecium, linked to difficult-to-treat infections. Mouse studies showed similar results. "Our study suggests reducing sweets intake during and after antibiotics could mitigate health risks," stated Jonas Schluter, PhD, at NYU Langone. Blood cancer patients were studied due to frequent antibiotic use and manageable food intake during hospitalization. "The microbiome is crucial in cancer care," noted Marcel van den Brink, MD, PhD, at City of Hope. (Science Daily)



Henri Kagan and Kenso Soai honoured for their work on asymmetric organic synthesis

Professors Henri Kagan and Kenso Soai have been awarded this year's prestigious prize by the Royal Swedish Academy of Sciences for their groundbreaking work in the field of chirality, sharing a 12 million Swedish kronor prize. Their journey began in 1985 when Professor Kagan discovered a method to manipulate chemical reactions, allowing for the creation of more of one mirror image than previously possible. Professor Soai furthered this in 1995 by designing the first chemical reaction with the potential to be homochiral, ultimately achieving a reaction where only one mirror image formed. This work is crucial as chirality is fundamental to life, affecting everything from drugs to biological systems. Dr. Phillip Broadwith highlights that mirror-image molecules can have vastly different biological effects, making this research vital. Professor Heiner Linke praised Kagan and Soai for solving a century-old mystery of spontaneous homochirality emergence, calling their work spectacular.



Scientists reprogram immune cells inside the body to fight cancer

CAR-T cell therapy has transformed the treatment of certain blood cancers but remains costly and slow, often inaccessible for many patients. Researchers at UC San Francisco have developed an innovative method that could overcome these challenges by reprogramming T cells directly inside the body, bypassing the need for extensive manufacturing. This breakthrough uses CRISPR-Cas9 gene-editing technology delivered by specialized particles targeting T cells, enabling them to produce cancer-fighting chimeric antigen receptors (CARs). Promising results in mice suggest this approach could streamline therapy, reduce costs, and enhance accessibility, offering hope for more efficient and equitable cancer treatment. This method, if successful in humans, could mark a significant advance in the field, potentially providing a faster, more affordable solution worldwide.



Flight passenger's odd appearance at major US airport leads officers to wild discovery

A traveler from Cuba, Alberto Hernandez Castillo, was apprehended at Miami International Airport after customs officers discovered 10 live birds hidden beneath his clothing. The birds, including yellow-faced grassquits and blue grosbeaks, were found inside cylindrical containers wrapped in tape and concealed within black pockets sewn into his underwear. Castillo, who is a legal U.S. resident, had denied bringing animals into the country on his customs declaration. However, upon inspection, officers noticed a suspicious bulge and conducted a pat-down that revealed the concealed birds. Charged with smuggling, Castillo claimed the birds were for personal use but failed to provide the necessary import documentation. He is set to appear in court on October 26.



Pomegranate compound improves heart function by up to 80% in study

A recent study by researchers at King's College London has uncovered promising results regarding urolithin A, a natural compound produced by the body after consuming foods like pomegranates, walnuts, and certain berries. This compound has shown the potential to enhance heart function significantly, by up to 80% in experimental models, particularly in cases of heart failure with preserved ejection fraction (HFpEF). This condition, affecting nearly half a million people in the UK, is characterized by the heart's inability to relax properly and fill with blood, despite maintaining effective blood-pumping capabilities. The symptoms of HFpEF, such as shortness of breath and fatigue, severely impact the quality of life. Current treatment options are limited and often focus on addressing underlying causes like aging, high blood pressure, and diabetes. Dr. Joseph Burgoyne, the senior author of the study, highlights the growing prevalence of this condition due to aging populations and increasing rates of obesity and diabetes. The discovery of urolithin A's benefits offers a glimmer of hope for more effective management of this challenging form of heart failure.



Jokes

Tantalum and Cobalt went on a date. They ate tacos.

Link to previous issues:

Membrane Transport

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Apple Maple Steamed Buns



Ingredients

- - 200g all-purpose flour
- - 100ml warm water
- - 5g active dry yeast
- - 2 medium apples
- - 50ml maple syrup
- - 10g cornstarch
- - 5g cinnamon powder
- - 1 pinch of salt

1. In a bowl, dissolve yeast in warm water and let it sit for 5 minutes until foamy. Add flour and a pinch of salt, then knead for 8-10 minutes until a smooth dough forms.
2. Cover the dough and let it rise in a warm place for 1-1.5 hours, or until doubled in size.
3. Peel, core, and finely dice the apples. In a small saucepan, combine diced apples, maple syrup, cornstarch, and cinnamon.
4. Cook the apple mixture over medium heat for 5-7 minutes, stirring occasionally, until apples are tender and the sauce has thickened. Let it cool completely.
5. Punch down the risen dough and divide it into 8 equal portions. Roll each portion into a flat circle.
6. Place a spoonful of the cooled apple filling in the center of each dough circle. Gather the edges of the dough and pinch them together to seal, forming a bun.
7. Place the buns on individual squares of parchment paper. Arrange them in a steamer basket, ensuring there is space between each bun.
8. Cover the steamer and let the buns rest for another 15-20 minutes for a second proof.
9. Steam the buns over boiling water for 12-15 minutes, or until they are puffed and cooked through.
10. Serve the Apple Maple Steamed Buns warm.